JON MORGAN

Technical Director – Building Performance & Sustainability

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Sustainability and systems professional with 30 years spanning building performance, ESD, and software development. Led technical analysis and simulation for a global engineering consultancy, co-directed an independent consulting practice, and built tools that turn complex analysis into decisions people can use. Most recently, led performance and sustainability initiatives for councils and developers across strategy, technical delivery, and decision support, with a growing focus on climate risk and adaptation.

Currently focused on director-level work in climate resilience and adaptive systems — informed by study of resilience engineering theory and a career-long recognition that complex systems require adaptive response, not prediction alone.

CORE COMPETENCIES

- Climate Resilience & Adaptation — deep technical grounding in heat and wind-driven building physics: heatwave stress-testing, thermal comfort modelling, and CFD/wind-tunnel analysis feeding into major architectural and structural design decisions.
- Regulatory & Rating-Framework Delivery — NCC Part J, Green Star, NABERS, BESS and JV3 compliance and certification delivered commercially at scale.
- Commercial & Operational Ownership — co-ran an independent ESD consulting practice for nearly a decade, closely involved in service-line profitability, cost management, strategy, branding and team leadership.
- Systems Architecture & Full-Stack Development — architects and builds custom software end-to-end, spanning UI, data pipelines, containerisation and local-first DevOps.
- Systems Engineering & Diagnostic Modelling — aerospace/defence-trained systems engineer; applies formal modelling method (abstraction, parameterisation, verification, validation) across aerospace, offshore, aviation, fire-safety and building-physics domains.

PROFESSIONAL EXPERIENCE

Co-Perform — Principal Jul 2018 - Aug 2026, Melbourne

Co-directed an independent ESD and building performance consultancy, delivering strategic advisory and technical work for councils, architects, builders and developers. Advisory work included sustainable building standards and heatwave-refuge strategy for the City of Melbourne, battery-storage sizing and payback for councils, and early design advice that let design teams select compliant façades quickly. Delivery spanned NCC Part J performance solutions (JV3), NatHERS rating delivery and QA from single dwellings to a 70-storey tower, NABERS Commitment Agreement modelling and Green Star upfront carbon submissions. Co-led the practice's business strategy and rebrand, owned its full analysis stack and built its reporting platform end-to-end.

Arup — Senior Consulting Engineer Sep 2005 - Aug 2014, Melbourne

Held the formal title of Australasian Building Physics Lead, with two direct reports and informal technical influence across a wider group of analysts embedded in other teams throughout APAC; owned the practice's simulation infrastructure and developed bespoke software and analytical workflows informing building performance decisions on major infrastructure and commercial developments.

Moreland Energy Foundation — Senior Consultant Sep 2014 - Jul 2018, Melbourne

Senior Consultant at Moreland Energy Foundation, advising on passive design, energy-efficient technologies, rooftop solar and storage integration — building custom engineering simulations, software tools and data visualisations. Also served as an Advisory Board Member for the Building Designers Association of Victoria's Energy & Sustainability Hub (2015–2018). Concurrent with running Visergy.

Visergy — Principal / Sole Trader Sep 2014 - Jul 2018, Melbourne

Operated as an independent sole trader under Visergy, delivering performance-based design support for town planning and building approvals (energy, daylight and thermal simulation) as a sub-consultant to ESD engineering firms — concurrent with a Senior Consultant role at Moreland Energy Foundation.

Early Career — DSTO, Scott Wilson Irwin Johnston, BMT Fluid Mechanics 1996 - 2004, Melbourne / Teddington, UK

Aerospace and defence engineering at DSTO, then building physics and fluid mechanics consulting at Scott Wilson Irwin Johnston and BMT Fluid Mechanics: energy, daylight and ventilation simulation, CFD, wind tunnel testing and offshore blast modelling.

SELECTED PROJECT HIGHLIGHTS

- Climate-Adapted Public Housing (Victorian DHHS) — thermal modelling lead in an MEFL-led consortium with DesignInc, stress-testing heat-resilient public housing for vulnerable tenants against spliced BOM heatwave records; three prototype units are now built and occupied at Mt Martha.
- Future Climate Weather File Application & Stress Testing — applied synthetic future-climate weather files to dynamic energy models, quantifying HVAC load and overheating risk in free-running buildings.
- City of Melbourne Resilient Buildings & Heatwave Refuges — lead technical sub-consultant (via Nation Partners) setting building standards, retrofit protocols and heatwave-refuge criteria so council buildings withstand bushfire smoke and extreme heat.
- Enterprise Reporting & Documentation Engine — built the pipeline producing all the practice's client reports: a single-source project database feeding automated, containerised document builds across ~10 report types.
- Richmond Window Film Retrofit — modelling showed an external window film would resolve an overheating problem exposing the owner to occupant legal claims, securing planning approval for a multi-hundred-thousand-dollar retrofit.
- Battery Storage Sizing Tool for Local Government — wrapped NREL's SAM engine in a containerised Python app to right-size solar-battery systems and test payback against commercial tariffs.
- Public Housing Retrofit Audit Platform (Victorian DHHS) — built a mobile audit platform combining building condition, tenant experience and billing data with in-home temperature logging to prioritise state-funded retrofits.
- Physics-Informed ML Surrogate Model (prototype) — trained regression models on hundreds of multi-zone thermal simulations to predict dynamic building performance in milliseconds.
- 2.5D-to-3D Computational Geometry Engine — built a procedural pipeline turning 2.5D building plan definitions into watertight, simulation-ready 3D geometry near-instantaneously, without reliance on full 3D CAD workflows.
- Victorian Local Government Capacity Building (MAV/CASBE) — designed and delivered 2025 training for council sustainability planners on auditing non-residential energy-modelling submissions, and helped test and refine the BESS framework.

TECHNICAL & DOMAIN EXPERTISE

Climate Resilience & Adaptation · Regulatory & Rating Frameworks (Green Star, NABERS, NCC Part J, BESS, JV3) · Thermal Dynamics & Building Physics · Fluid Dynamics & Atmospheric Flow · Energy Systems & Infrastructure · Critical Infrastructure & Transport · Offshore & Marine Engineering

CREDENTIALS

- NABERS Upfront Carbon Accredited Assessor (2026)
- Advisory Board Member, Energy & Sustainability Hub — Building Designers Association of Victoria (2015–2018)

EDUCATION

- Master of Urban Planning and Environment — RMIT University (2009–2014)
- Bachelor of Engineering (Aerospace) — RMIT University (1991–1995)